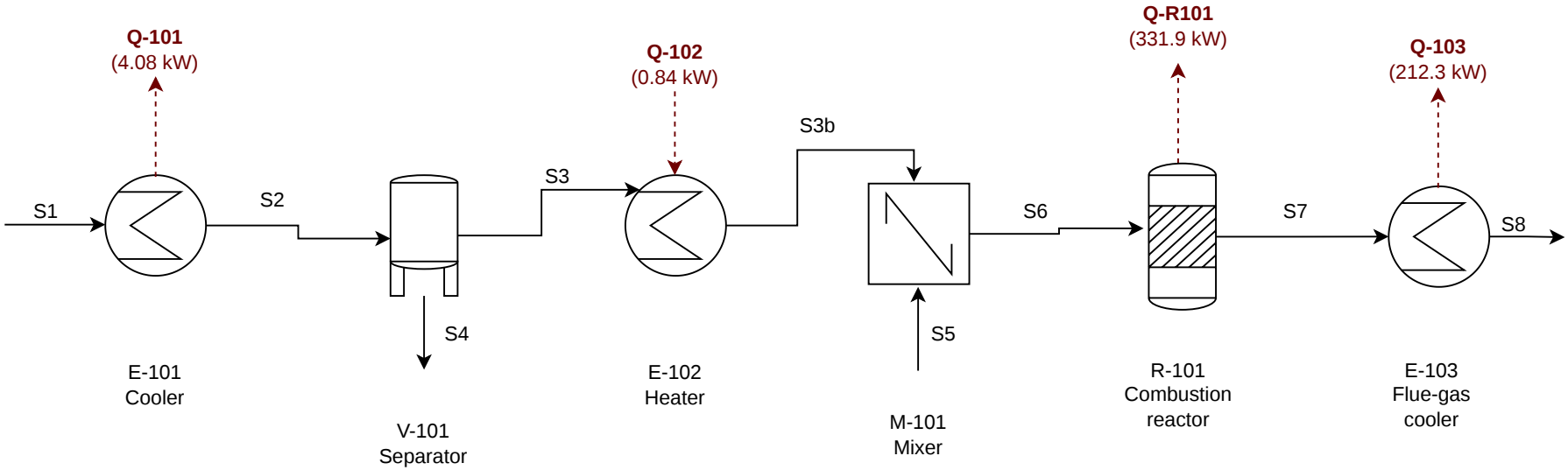


Stream	S1	S2	S3	S4	S3b	S5	S6	S7	S8
Description	*Biogas*	*Cooled gas*	*Dry gas*	*Condensate*	*Warmed gas*	*Air*	*Mixed gas*	*Hot flue gas*	*Cooled flue gas*
T (°C)	37	5	5	5	25	25	25.0	800	120
P (bar)	1.05	1.05	1.05	1.05	1.05	1.05	1.05	1.05	1.05
Phase	V	V+L	V	L	V	V	V	V	V
Molar flow (kmol/h)	4.46	4.46	4.221	0.239	4.221	28.30	32.52	32.52	32.52
x CH4	0.580	0.580	0.613	–	0.613	–	0.0795	0.0008	0.0008
x CO2	0.360	0.360	0.380	–	0.380	–	0.0494	0.1281	0.1281
x H2O	0.060	0.060	0.007	≈1	0.007	–	0.0009	0.1584	0.1584
x O2	–	–	–	–	–	0.21	0.1827	0.0252	0.0252
x N2	–	–	–	–	–	0.79	0.6875	0.6875	0.6875

Material stream

Energy stream (heat duty)

E = heat exchanger

V = vessel

M = mixer

R = reactor

PFD – Biogas upgrading and combustion

Process simulation in DWSIM 10.2.11 (Peng-Robinson)
Basis: 4.46 kmol/h biogas (58 % CH4), 99 % CH4 conversion at 800 °C

Prepared by: Louay Ouhibi Date: 27/10/2026